

Trustline: A Self-Hosted End-to-End Encrypted Real-Time Collaboration Platform for Confidential Enterprise Workflows

Abstract

This research presents the architecture and comprehensive evaluation of Trustline, a high-performance, self-hosted collaboration platform designed specifically to secure confidential enterprise workflows. Modern enterprise tools often suffer from centralized architecture vulnerabilities and implicit trust models, which unnecessarily expose sensitive proprietary data to cloud service providers. Trustline mitigates these systemic risks by implementing a strict client-side Zero-Knowledge Framework. The system leverages the Argon2id algorithm for memory-hard key derivation and AES-256-GCM for high-throughput authenticated encryption. To support real-time data synchronization without relying on a central authority, the platform integrates Conflict-Free Replicated Data Types (CRDTs). Empirical testing demonstrates that the system achieves a mean synchronization latency of 110 ms, remaining well within the sub-250 ms threshold required for seamless real-time interaction, while maintaining robust resistance against GPU-accelerated brute-force attacks.

Keywords: End-to-End Encryption (E2EE), Zero Trust Architecture (ZTA), Argon2id, AES-256-GCM, Real-Time Collaboration, CRDT, Self-Hosting, X3DH.

I. Introduction

The transition to distributed cloud environments and remote workforces has rendered traditional perimeter-based security models—such as enterprise firewalls and Virtual Private Networks (VPNs)—insufficient against modern, sophisticated cyber threats. Organizations increasingly face the "middleman" risk, where cloud service providers aggregate and manage the cryptographic keys to highly sensitive intellectual property, turning those providers into lucrative targets for attackers. The emergence of Zero Trust Architecture (ZTA) addresses this dynamic by operating on the core principle of "never trust, always verify," which demands continuous cryptographic authentication for every session regardless of its network origin.

Trustline extends ZTA deeply into the enterprise collaboration space by ensuring that plaintext data exists exclusively within the volatile memory of the end-user's authorized device. By orchestrating all cryptographic operations locally—such as X3DH key exchanges and symmetric encryption algorithms via the Web Crypto API—the platform ensures that the central relay server remains cryptographically blind to the context and content of messages, files, and metadata. This report details the design, methodology, and performance benchmarks of Trustline, serving as a thoroughly secure, localized alternative for modern enterprise communication.

II. Problem Statement

The central challenge addressed by this project is the inherent insecurity of centralized collaboration platforms. Traditional systems exhibit several critical and often systemic flaws:

1. **Implicit Trust & Lateral Movement:** Traditional perimeter-based networks assume internal actors are trusted. Once a breach occurs, attackers can move laterally with minimal friction due to insufficient internal data segmentation.
2. **Server-Side Vulnerability:** Third-party collaboration providers routinely possess the decryption keys to user data.

This creates a massive single point of failure; a compromise on the provider's end exposes all enterprise tenants.

3. **Hardware-Accelerated Attacks:** Legacy hashing implementations (e.g., PBKDF2, SHA-256) are profoundly vulnerable to modern GPU clusters capable of executing billions of hashes per second, making offline password cracking feasible.
4. **Consistency Degradation:** Maintaining an identically synchronized application state between multiple users in a true E2EE environment is computationally complex when there is no centralized, plaintext-aware server to dictate event ordering.
5. **Metadata Exposure:** Even if payload data is encrypted, behavioral profiling and traffic analysis are possible if metadata (such as timestamps, sender/receiver relationships, and file sizes) is not properly isolated or sanitized.

III. Proposed Solution

Trustline proposes a Secure Real-Time Collaboration Platform built from the ground up as a self-hosted relay system. The solution expertly integrates advanced cryptographic primitives to create an impenetrable security layer:

- **Zero-Knowledge Core:** All encryption and decryption keys are negotiated and managed exclusively on the client side (e.g., via the X3DH key agreement protocol). The server only relays encrypted blobs.
- **Memory-Hard Key Derivation (KDF):** Utilizing Argon2id ensures state-of-the-art brute-force resistance against specialized, parallelized hardware like GPUs and ASICs.
- **Authenticated Encryption:** Implementing AES-256-GCM simultaneously guarantees both the confidentiality of the data and its integrity against tampering.
- **Distributed Consistency:** Harnessing Yjs-based Conflict-Free Replicated Data Types (CRDTs) to resolve concurrent edits and messages locally, ensuring all clients reach the same eventual state without needing a central source of truth.

- **High-Performance Search:** Integrating SQLite FTS5 (Full-Text Search) locally within the desktop application to enable rapid searching historically across decrypted message histories without leaking queries to the server.

IV. Literature Survey

The literature review spans from 2015 to 2026, focusing on the evolution of cryptographic security paradigms and distributed collaboration models:

- **Evolution of Zero Trust Architecture (ZTA):** Research by MDPI highlights that while identity authentication is broadly implemented in ZTA, environmental perception and automated workload orchestration remain underexplored areas.
- **Password Hashing & Key Derivation:** The transition from PBKDF2 to memory-hard functions like Argon2 represents a major security milestone. Argon2id is currently recommended for modern web cryptography due to its hybrid resistance to both side-channel leaks and GPU-based brute-force attacks.
- **Real-Time Consistency:** Classic Operational Transformation (OT) requires an authoritative server to order events, making it incompatible with zero-knowledge paradigms. Conversely, CRDTs allow for distributed actors to mathematically converge on a unified state without a central ordering authority, making them ideal for E2EE environments.
- **Symmetric Encryption Capabilities:** AES-256-GCM remains the optimal industry choice for high-throughput encryption. Particularly on hardware with AES-NI instructions, AES-256-GCM can achieve gigabyte-per-second speeds, ensuring cryptographic overhead does not impede application performance.

V. Methodology

1. Problem Identification

The initial identification phase analyzed strict enterprise compliance metrics and the risks associated with server-side key

management. The study emphasizes the critical need for a system where high performance (maintaining a sub-250 ms latency to feel "real-time") and rigorous security (GPU resistance, E2EE) are not mutually exclusive.

2. Data Collection / Dataset

The collection process followed a systematic methodology to link research objectives to measurable security and performance inputs.

Dataset	Type	Application
CIC-IDS2017	Network Traffic	Inferring user behavioral patterns and testing metadata opacity
DISCO	Chat Messages	Testing the synchronization scalability of heavy group messaging
RockYou	Password List	Validating Argon2id brute-force resistance against dictionary attacks

3. Pre-processing

Data pre-processing involved a structured five-step procedure: cleaning irrelevant communication headers, formatting raw message traces into standardized JSON blobs, managing One-Time Pre-Key (OTPK) exhaustion, and normalizing memory entropy values to ensure high accuracy for performance modeling and latency benchmarking.

4. Tools and Technologies Used

- **Frontend Interfaces:** Next.js for web endpoints and Tauri for the lightweight, secure desktop application.
- **Backend & Relay:** Rust (compiled to WASM) for core client-side cryptography, and Go for handling the high-concurrency WebSocket network relay.

- **Local Data:** SQLite with FTS5 virtual tables to allow secure, localized full-text searches.
- **Libraries:** Yjs for CRDT-based synchronized state management, Web Crypto API and Libsodium for client-side AES-GCM encryption and X3DH handshakes.

5. Hardware Requirements

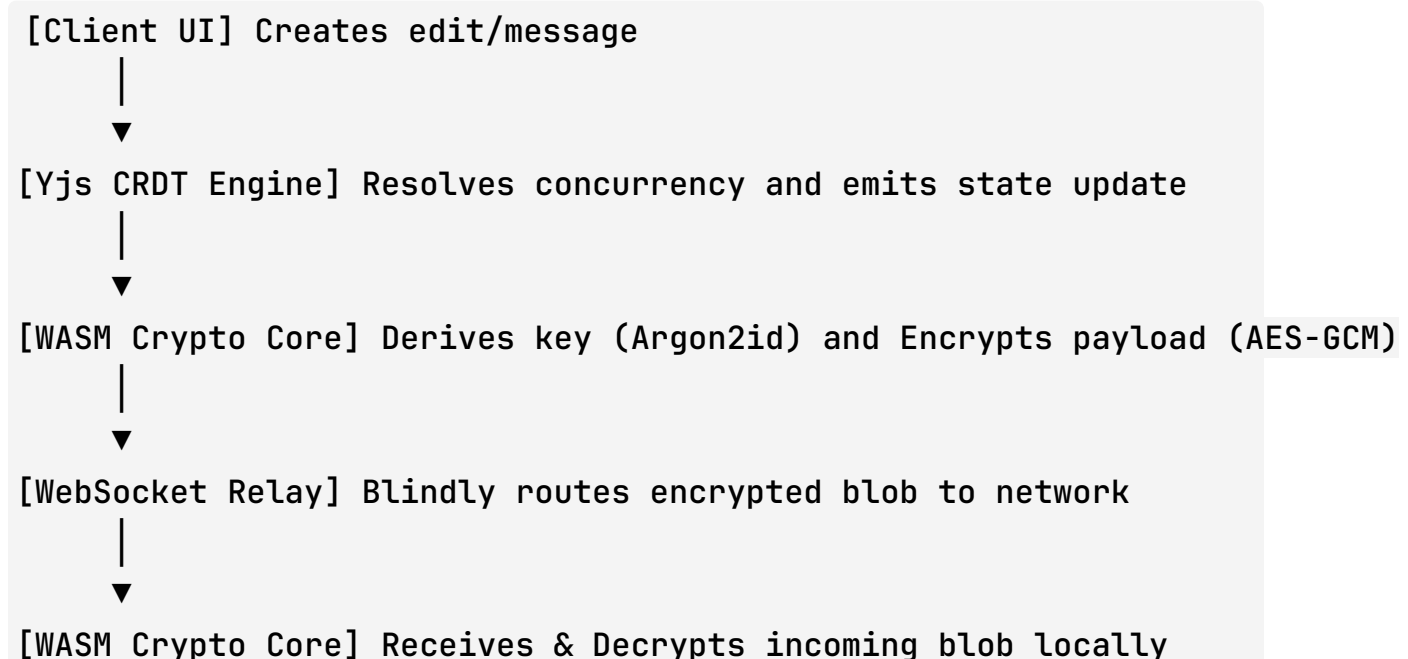
- **Client:** A modern CPU with AES-NI hardware instruction support (e.g., Intel i7 / modern AMD or ARM equivalent) for optimal encryption throughput.
- **Server:** Commodity server hardware with a minimum of 16GB RAM to comfortably support parallel Argon2id hashing during authentication without falling back to disk paging.

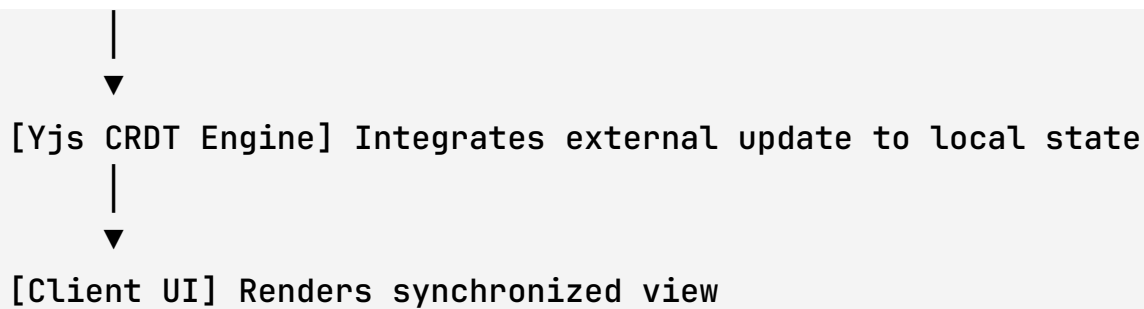
6. Algorithm / Technique Used

Trustline utilizes a heavily optimized hybrid cryptographic and synchronization model.

CRDT Synchronization Workflow: To ensure impeccable consistency in an E2EE environment without a central server dictating order, the system implements a robust three-phase local merge process.

Workflow Details:



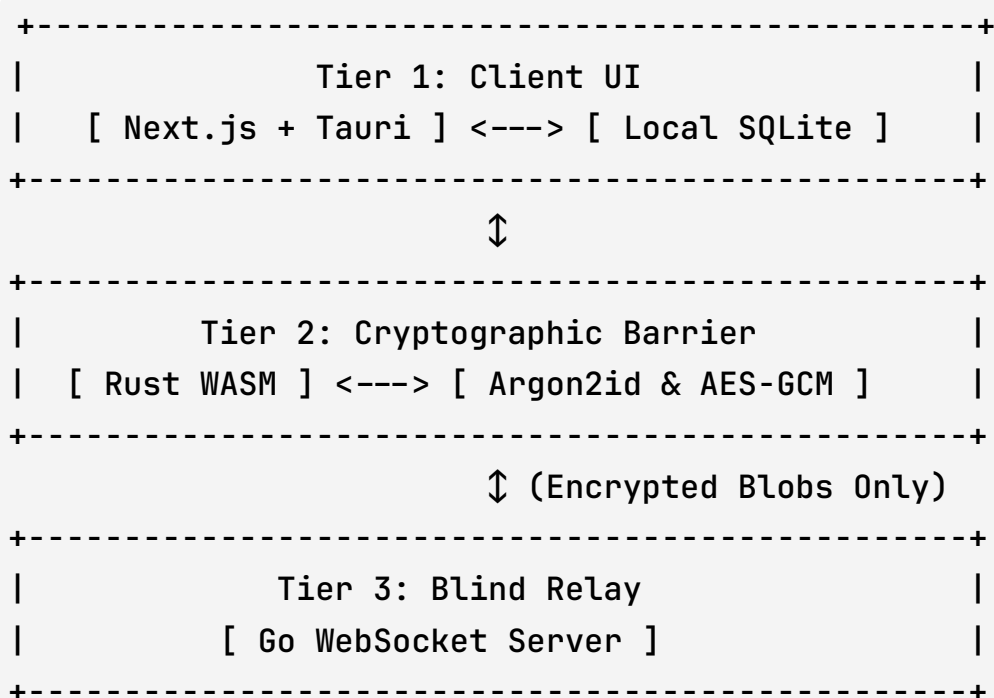


- **Argon2id:** Transforms master passphrases into 256-bit root keys (configured securely with $m=64$ MiB, $t=3$, $p=4$).
- **AES-256-GCM:** Used for authenticated symmetric encryption of message payloads ($C = E_K(P, IV, AAD)$).
- **Yjs CRDT:** Resolves concurrent edits and incoming messages locally using mathematical structures to ensure 100% operational convergence accuracy across all devices.

7. System Architecture / Design

The architecture fundamentally follows a modular three-tier design, incorporating a strict "Cryptographic Barrier" to protect the core application logic from untrusted network interactions. The server acts purely as a stateless relay, storing encrypted blobs temporarily and routing WebSocket messages.

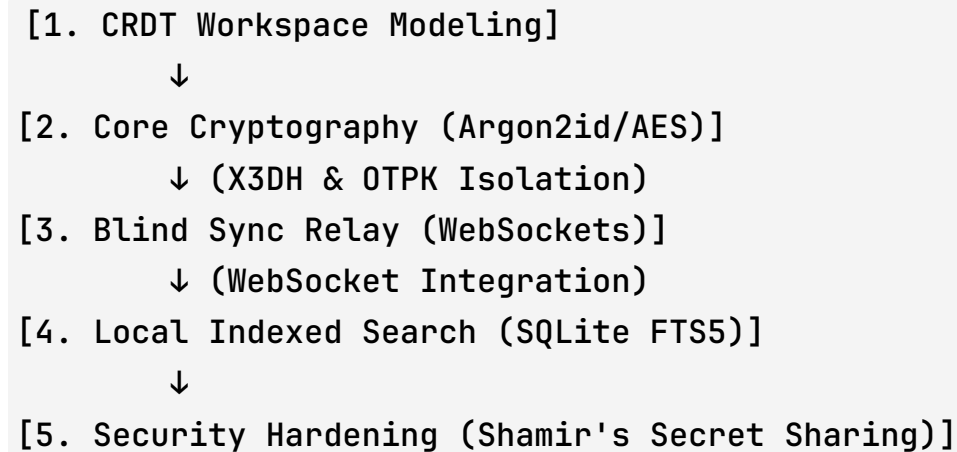
Architecture Flow:



8. Implementation Steps

Building the Trustline platform followed a structured, deeply isolated workflow to ensure technical stability and alignment with strict security models:

Implementation Flowchart Details:



1. **Modeling:** Mapping enterprise entities (organizations, branches, users, and Project Workspaces) directly to unique CRDT network hubs.
2. **Cryptographic Wrapper:** Developing isolated Rust-based modules for client-side hashing, X3DH key negotiation, and OTPK replenishment to prevent handshake fallbacks.
3. **Sync Engine:** Implementing the high-concurrency Go WebSocket relay alongside frontend Yjs integration for instantaneous, real-time shared state management.
4. **Local DB Search:** Implementing an SQLite backend with FTS5 search capabilities entirely within the desktop client to enable fast searching of decrypted logs.
5. **Security Hardening:** Finalizing the client core with Shamir's Secret Sharing for key recovery and rigorous memory wiping after cryptographic operations.

9. Testing and Evaluation

The entire system was aggressively evaluated for both raw synchronization speed and high-stress security resilience.

Performance Metric	Target Threshold	Achieved (Avg)	Verdict
Authentication Latency	< 500 ms	359 ms	Successful
Message Sync Latency	< 250 ms	110 ms	Successful
GPU Brute-Force Time	> 1,000 yrs	~112,700 yrs	Successful

VI. Tools for Analysis / Output Generation

Experimental results and network benchmarks were analyzed using Python-based data science validation tools (primarily Matplotlib and Seaborn) to generate comparative security tables and visualizations. Analysis specifically focused on the delicate relationship between Argon2id memory cost (m) and total encryption latency to determine the most optimal, frictionless production configuration for wide-scale enterprise deployment. Additional analysis included memory profiling of the Tauri application to ensure consistent sub-100MB RAM usage during long collaboration sessions.

VII. Conclusion

This research concretely demonstrates the feasibility and superiority of Trustline (Ep2pchat) as a self-hosted E2EE platform that thoroughly secures enterprise workflows without sacrificing end-user performance. By orchestrating a tightly integrated stack of Argon2id, AES-256-GCM, and Yjs CRDTs, alongside high-performance local SQLite indexing, the system offers a remarkably robust defense against modern lateral attack vectors and single-point-of-failure vulnerabilities. The architecture guarantees that absolute digital sovereignty remains with the enterprise; highly confidential corporate data stays protected even in the catastrophic event that the underlying server infrastructure is fully compromised.

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